

Cross-Constituency Aggregation for Community Notes

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Abstract

Community Notes is X’s crowd-based fact-checking system: contributors write contextual notes, other contributors rate them, and a bridging algorithm decides which notes become visible. The algorithm is designed to avoid simple majority capture by estimating note quality separately from ideological alignment, so a note should gain support across disagreement rather than only within one side [Wojcik et al., 2022, X Community Notes, 2026]. This design is powerful, but it also creates a visibility problem. Because the system works through sparse and unequal participation, display decisions can depend heavily on the ratings supplied by a relatively small active core, and many notes remain in Needs More Ratings when the algorithm does not find enough diverse support [Goyal et al., 2026, Razuvayevskaya et al., 2025]. We argue that this is not only a data-sparsity problem. It is also an aggregation problem: Community Notes treats cross-group agreement indirectly, through a fitted model, rather than as an explicit decision rule over constituencies.

We take inspiration from political science and social choice. In systems such as Switzerland’s double-majority referendum rule, important decisions must gain support not only from a national majority but also from a majority of cantons; the system treats cross-group support as part of the decision’s legitimacy. Our philosophy is simple: a note should not become visible because one active group overwhelms the rest; it should become visible when each relevant constituency has a real chance to support it. We operationalize this idea as cross-constituency aggregation. We recover behavioral constituencies from co-rating patterns among 200,000 active raters, compute each note’s approval rate inside each constituency, and combine those rates with a geometric mean under a minimum-coverage rule. In a 100,000-note analytical slice, Community Notes displays 6,832 of 44,722 representative-picked notes (15%). Our rule identifies a 13,655-note rescue pool, raising potential visibility to 20,487 notes (46%) before validation. A second-stage validation screen classifies 3,896 candidates as sourced, factual, and rescue-worthy. These results suggest that useful notes can remain hidden when cross-group support is modeled only indirectly.

1 Introduction

Community Notes is X’s crowdsourced fact-checking system. When users believe a post needs context, contributors can write a short note. Other contributors then rate whether that note is

helpful. If the system decides that the note has enough support, the note is shown under the post. This makes Community Notes a large public experiment in crowd-based moderation: the platform does not rely only on professional fact-checkers or automated classifiers, but asks a crowd to decide which context should become visible [Wojcik et al., 2022, X Community Notes, 2026].

The key innovation of Community Notes is its bridging algorithm. A simple majority vote would be easy to capture by one political side or one active group. X’s current 2026 official guide describes the main ranking layer as matrix factorization over the sparse note-rater matrix. In simplified form, the predicted rating is

$$\hat{r}_{un} = \mu + i_u + i_n + f_u \cdot f_n.$$

This decomposes a rating into four parts:

- μ : the global average rating level;
- i_u : the rater intercept, capturing whether a rater is generally more generous or stricter when rating notes;
- i_n : the note intercept, used as the note’s helpfulness score;
- $f_u \cdot f_n$: the match between the rater factor and the note factor, capturing whether a note appeals to raters with particular viewpoints.

The important point is that a note receives a high helpfulness score only when its support cannot be explained merely by viewpoint compatibility. X therefore presents the model as a way to identify notes with broad appeal across viewpoints [X Community Notes, 2026].

This matrix-factorization layer is not the whole current production system. The official ranking guide also describes agreement safeguards, Net Helpful minimums, checks for rater engagement and rater independence, topic and group models, and a November 2025 Gaussian final-scoring pilot that approximates a geometric mean of helpfulness rates across the rater-factor spectrum [X Community Notes, 2026]. We therefore treat the formula as the core ranking layer, not as the full Community Notes pipeline.

This design is powerful, but it also has a problem. After viewpoint agreement is modeled, the system still estimates note helpfulness from sparse and uneven ratings. Recent work argues that this makes the quality estimate sensitive to rater quality, noisy raters, strategic raters, and hyperactive minorities [Goyal et al., 2026, Nudo et al., 2026]. The official guide

itself reflects this concern: its rater-engagement safeguard re-computes a baseline intercept after omitting the top 0.1% of rating-volume contributors, in order to check whether notes are helpful beyond the most active raters [X Community Notes, 2026]. Participation in Community Notes is also highly unequal. Razuvayevskaya et al. report that the top 10% of contributors produce 58% of all notes, while most users remain much less active [Razuvayevskaya et al., 2025]. This matters because a crowd system can formally include many people while still depending heavily on a small, active core.

The reason is structural. Community Notes faces sparse and uneven participation: most raters see only a small part of the note universe, while a small number of users rate very often. The current model answers this problem as an estimation task. It adjusts for rater-level tendencies, estimates note helpfulness, and uses a latent rater-note factor to capture whether a note fits the raters who evaluate it. This is a reasonable way to prevent one-sided agreement from being mistaken for broad quality. But it still keeps group support inside a fitted model. It does not create direct quantities such as “approval inside constituency A” or “approval inside constituency B.” As a result, cross-group acceptance is present only indirectly, and a small active core can still shape which signals the model sees most often.

Our approach starts from a different premise about the disagreement itself. Latent-quality estimation makes sense only if rater disagreement is noise around a shared quality signal, something that washes out once it is partialled away. On a platform that works as a de facto public sphere, that picture does not hold. Raters do not fluctuate randomly around one true score; they cluster by what they attend to: which claims are worth checking, what counts as credible evidence, what context feels necessary. Those judgments cohere within a group and pull apart across groups. The disagreement that results is neither noise nor reducible to a single ideological axis. It is structured: persistent, correlated across issues, and organized into behavioral blocs that we later recover from real co-rating patterns.

Because the disagreement is structured, we do not try to decide which raters are better, more sincere, or more reliable as individuals. We treat the recovered constituencies as legitimate parts of the crowd and ask whether a note is acceptable across them, which turns Community Notes from a prediction problem into a crowd-agreement problem: the platform is making a collective decision, in the name of “the community,” about whether a note should be attached to a public post. The question this paper asks follows directly:

When ratings come from behaviorally distinct constituencies, which aggregation rule should decide whether a Community Note becomes visible?

We compare the platform’s current decision with a rule that requires meaningful support from each recovered constituency, treating disagreement as a feature of the governance problem rather than as noise to be averaged away. This paper makes three contributions:

- We reframe the Community Notes display decision as an

aggregation problem over behaviorally distinct constituencies, rather than a latent-quality estimate over a pooled crowd.

- We recover those constituencies endogenously from co-rating behavior, with no external party, country, or ideology labels, and give a transparent standing test for when a recovered cluster may count as a constituency.
- We operationalize cross-group legitimacy as a concrete non-compensatory rule (a geometric mean under minimum coverage) and show, on a 100,000-note slice, that it surfaces a large pool of otherwise-hidden public-interest notes, which a second-stage screen validates as sourced and factual.

The inspiration is political science and social choice rather than another rater-quality model. Divided societies often use power sharing, double majorities, parallel consent, vote pooling, or veto-like rules so that collective decisions depend on support traveling across groups, not only on the strongest or most active side [Lijphart, 1977, McGarry and O’Leary, 2009, Horowitz, 1991, Reilly, 2002]. Social-choice theory gives a formal version of the same idea: legitimate collective decisions should represent relevant parts of the public, not only the largest or most active part [Peters, 2018, Qiu, 2025]. We borrow this logic in a limited way. Community Notes is not a state, and displaying a note is not passing a law. Still, both settings face the same aggregation problem under disagreement.

Philosophy

Our philosophy is simple: all recovered constituencies count. A note should become visible when it can survive cross-group acceptance, not when one active group dominates the observed rating record. Over time, such a rule may also encourage contributors to write and rate notes that can travel across camps, because visibility depends on cross-constituency support rather than one-sided enthusiasm.

Principles

P1. Presence. Every constituency must be present in the decision.

P2. Non-compensation. One constituency’s rejection should not be fully compensated by another constituency’s enthusiasm.

P3. Symmetry. Constituencies should enter the rule symmetrically, so that no group is privileged by default.

P4. Behavioral recovery. Constituencies should be recovered from observed rating behavior rather than assigned from outside labels such as party, country, or ideology.

We operationalize these rules as cross-constituency aggregation. Using the active 200k Representative pipeline, we cluster active raters from co-rating behavior, reassign them into two main behavioral constituencies, compute each note’s

approval rate inside each constituency, and combine these rates with a geometric mean under a minimum-coverage requirement. The geometric mean creates a soft veto: if either constituency strongly rejects a note, the final score falls, even if the other constituency is highly supportive. We then use this score to identify hidden Community Notes that deserve a second look, and we validate those candidates separately for factuality, sourcing, relevance, and usefulness.

Concretely, let a_{ic} be note i 's approval rate inside constituency c . Writing H_{ic} and N_{ic} for the helpful and total ratings the note receives from constituency c ,

$$a_{ic} = \begin{cases} H_{ic}/N_{ic}, & N_{ic} > 0, \\ 0.5, & N_{ic} = 0, \end{cases}$$

where the 0.5 fallback marks abstention rather than endorsement: a constituency that has never rated the note casts no vote. The cross-constituency score is the geometric mean

$$C_i = \sqrt{a_{i1} a_{i2}},$$

and a note is selected when $C_i \geq 0.5$ and each constituency contributes at least three raters; when Community Notes does not already display such a note, it enters the rescue pool. Because C_i falls toward zero as either approval rate does, strong support from one constituency cannot rescue a note the other rejects (P2), while the rule reads only within-constituency rates, so a larger or busier bloc earns no extra weight (P3). The minimum-coverage requirement enforces presence (P1), and the constituencies themselves are recovered from co-rating behavior rather than assigned from outside labels (P4).

In a 100,000-note analytical slice, Community Notes displays 6,832 of 44,722 representative-picked notes, or 15%. Cross-constituency aggregation identifies a 13,655-note rescue pool, raising potential visibility to 20,487 notes, or 46%, before validation. A second-stage validation screen classifies 3,896 of these candidates as sourced, factual, and rescue-worthy. These results suggest that Community Notes may have an aggregation problem as well as a rating problem: useful public-interest notes can remain hidden when cross-group support is modeled only indirectly.

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